Certifying Retrieval-Policy Deployment with Non-Neutral AI Judges: Decision Weights, Control Variates, and What Judge Accuracy Does Not Tell You

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Abstract

Before a retrieval policy is deployed in a new domain, an auditor must certify, from a limited human audit, that it is not worse than the alternatives by more than a tolerance ε. Large language models (LLMs) can now judge relevance at scale, which raises a practical question: *which* AI judge helps the human audit, by how much, and can the certificate still be trusted when the judge is biased toward one of the policies it is judging? We study this question for menus of cutoff policies over a shared ranked candidate pool, with set-F1 or precision utilities, on eight retrieval collections, five AI judges, and a pre-registered external evaluation. Three results follow. (i) The audit savings obtainable from a judge are governed not by its label accuracy but by the correlation ρ between true and judge-predicted *paired policy differences*: over 54 (collection, policy-pair, judge) settings the effective-sample-size gain of prediction-powered inference correlates 0.78 with ρ and -0.02 with accuracy, the relation holds inside every collection, and a judge built from the same model that defines a policy over-credits that policy by 0.14 F1 on paired differences. A mechanism map shows that savings appear only when $\rho \gtrsim 0.55$, the unlabeled pool is at least twice the audit, and the decision is hard. (ii) A sample-split certificate remains valid for *any* judge, including an adversarial one, because the judge enters only as a control variate; we measure wrong-certificate rates ≤ 0.008 across 24 development cells and 0/500 or $\leq$ 6/500 in three pre-registered evaluations, and we show that the earlier in-sample certificate exceeded its nominal risk (0.146 and 0.180 at $\alpha = 0.10$). (iii) Defining the estimand precisely yields closed-form per-document *decision weights*; importance sampling by these weights cuts the human judgments needed to certify by more than half relative to uniform document sampling, a judge used as a document-level control variate adds a further 14–21% when it is good and nothing when it is poor, and a faithful active-inference rule with a pilot-estimated residual model matches this within noise, whereas the calibrated-judge rule is 30–50% worse. We claim no new estimation theory: the contribution is the exact mapping of policy certification onto existing inference tools, the conditions under which AI judges pay, and honest reporting of three pre-registered evaluations in which validity replicated every time while the pre-registered efficiency criteria did not: once because the judge's ρ and the unlabeled pool were too small, once because the locked budget grid was too small for the collection, and once because it was too large and every method saturated—while the co-registered cost metric halved with decision-weighted sampling and fell a further 26% with a judge whose pair accuracy was only 0.47.

1 Introduction

Retrieval systems are increasingly deployed as configurable policies: a fixed ranker plus a rule that decides how many documents to return, which threshold to apply, or which reranker to trust. When such a system moves to a new domain, an auditor with a small human-judgment budget must decide whether a candidate policy can be deployed—formally, whether its utility is within ε of the best alternative on the target distribution—or whether to collect more judgments or abstain. Two developments make this problem timely. First, LLM relevance judges are cheap enough to label every candidate document, so the audit can be assisted by a judge. Second, the policies themselves are often built from the same model families as the judges, so a judge may not be neutral toward the policies it evaluates (Balog et al., 2025).

Prediction-powered inference (PPI) and active statistical inference provide the statistical machinery for combining AI predictions with human labels (Angelopoulos et al., 2023a;b; Zrnic & Candès, 2024). Applied naively, however, they leave three questions open that an auditor must answer before spending a budget: *which* judge to use, *whether* the resulting certificate can be trusted if the judge is wrong or biased, and *where* to spend the human labels. This paper answers all three for the policy-certification problem, and reports what happened when the answers were tested on data that had been locked before analysis.

Contributions.

1. **What a judge is worth is not its accuracy.** For paired policy differences, the PPI effective-sample-size gain is $1/(1-\rho^2)$ where ρ is the correlation of true and predicted differences (Proposition 1). Across 54 (collection, pair, judge) settings this gain correlates 0.78 with ρ and -0.02 with pair-level judge accuracy; the relation holds within every collection and for judge swaps at fixed policies (Section 3). A mechanism map on real data (Figure 2) shows the gain materialises only when $\rho \gtrsim 0.55$, the unlabeled pool is at least twice the audit, and the decision is hard. A judge built from the model that defines a policy over-credits that policy by 0.14 on paired F1 and its ρ for that policy's comparisons collapses from 0.57 to 0.25 (Section 3.1).

2. **A certificate that is valid for any judge.** Splitting the audit into a training half (policy parameters, candidate, judge adoption) and a validation half (the bound) makes the certificate valid conditionally on every choice made on the training half, for any judge, because the judge enters only through an unbiased control variate (Proposition 2). We show empirically that the earlier in-sample certificate exceeded its nominal risk, that the split certificate stays at ≤ 0.008 wrong certificates across 24 cells and at 0/500 under an adversarial judge, and that an *always-use-the-judge* policy can lose efficiency with a poor judge while a pilot-based adoption rule protects it (Section 4).

3. **Where to spend human judgments.** For utilities with a label-independent normaliser, the paired difference is a linear functional with closed-form per-document decision weights (Proposition 3); the common documents carry weight $|1/k_a - 1/k_b|$, so a band-only audit is exact only when cutoffs coincide. Importance sampling by these weights, with the judge as a document-level control variate, reduces the human judgments needed to certify by more than half relative to uniform sampling; the judge adds 14–21% when good and nothing when poor; a faithful active-inference rule matches this, and a calibrated-judge rule is 30–50% worse (Section 5). For set-F1 we give a bias-corrected linearisation and measure its remainder (Appendix A).

4. **Three pre-registered external evaluations.** On TREC Deep Learning 2021–2023 (211 queries) the query-level methods, on TREC CAsT 2019 (159 conversational turns) the document-level methods, and on ANTIQUE (200 non-factoid questions) the λ-corrected control variate were evaluated under locks committed before the data were touched. Validity replicated every time (wrong certificates 0/500, 0/500 and $\leq 6/500$; boundary type-I error ≤ 0.013). The pre-registered efficiency criteria were not met in any of the three, each time for a reason we can name—too small a ρ and unlabeled pool, too small a budget grid, a saturated one—and we report all three as is (Section 7). The CAsT evaluation exposed a defect of the coefficient-1 control variate under an adversarial judge; the ANTIQUE evaluation confirmed the λ-corrected fix.

What we do not claim. We do not propose a new estimator or a new sampling optimality result; our estimator is a special case of active inference for a linear functional (Section 5.1). We do not claim that AI judges reduce audit cost unconditionally; the pre-registered evaluation shows they need not. We do not claim a general theory of judge neutrality; the self-preference signature we measure is specific to the identical-model judge.

2 Setup

Let P be the target query distribution, $\mathcal{M}$ a finite menu of policies acting on a shared ranked candidate pool for each query q, and $u_m(q) \in [0,1]$ the utility of policy m on q given human relevance labels $y_d \in \{0,1\}$ for the pooled documents d. Two utilities are used: set-F1, $u_m(q) = 2\,\mathrm{tp}_m/(k_m + n_G)$ with $k_m = |R_m(q)|$ and $n_G(q) = \sum_d y_d$, and precision at the policy's cutoff, $u_m(q) = \mathrm{tp}_m/k_m$. Population utilities are $\mu_m = \mathbb{E}_P[u_m(q)]$.

Decision and error. A certificate ACT selects a candidate $\hat{m}$ and asserts $\max_j \mu_j - \mu_{\hat{m}} \le \varepsilon$. A *wrong certificate* is an ACT whose true regret exceeds ε. All methods use the same risk budget: for looks $t = 1..L$ and ordered pairs $(j, \hat{m})$, each one-sided upper confidence bound $U_{t,j}$ on $\mu_j - \mu_{\hat{m}}$ is at level $1-\alpha'$ with $\alpha' = \alpha/(L\,|\mathcal{M}|(|\mathcal{M}|-1))$, and $\mathrm{ACT}_t \iff \max_{j\neq\hat{m}} U_{t,j} \le \varepsilon$. Throughout $\alpha = 0.10$; $\varepsilon \in \{0.005, 0.01, 0.02, 0.05\}$.

Audit units and the judge. A *query-level* audit labels a query's whole pool (needed for set-F1 because of n_G); a *document-level* audit labels a sampled subset of documents. An AI judge is a fixed map f from a (query, document) pair to a relevance probability; we threshold it at 0.5 for judge utilities $\hat{u}_m(q)$ and use the probability where a rule requires it. Nothing below assumes the judge is calibrated, unbiased, or independent of the policies.

Efficiency metric. J_{50} is the number of human-judged (query, document) pairs at which the cumulative ACT rate first reaches 50%, including every human label consumed (pilot/training queries, judge selection, residual models, and the bound itself). Between budgets actually run we interpolate linearly; a cell whose largest budget stays below 50% is reported as *not reached* and never enters a ratio. Wrong-certificate rates are reported next to every J_{50} with Clopper–Pearson 95% intervals. Query-level (set-F1) and document-level (precision) blocks use different utilities and are never compared to each other.

3 What determines the value of a judge

Proposition 1 (PPI gain for paired differences)**.** *Fix policies j, m and let $D = u_j - u_m$, $\hat{D} = \hat{u}_j - \hat{u}_m$ with variances $\sigma^2, \hat{\sigma}^2$ and correlation ρ. With n audited and $N \gg n$ unlabeled queries, the PPI++ estimator $\hat{\mu}(\lambda) = \lambda\,\overline{\hat{D}}_N + \overline{(D - \lambda\hat{D})}_n$ has variance $[\sigma^2 - 2\lambda\rho\sigma\hat{\sigma} + \lambda^2\hat{\sigma}^2]/n + \lambda^2\hat{\sigma}^2/N$, minimised at $\lambda^\star = \rho\sigma/\hat{\sigma}$ to $\sigma^2(1-\rho^2)/n$. The effective-sample-size gain over the human-only estimator is therefore $1/(1-\rho^2)$: the judge enters only through ρ.*

The proof is a direct variance computation (Appendix B). Writing $\hat{u}_m = u_m + b_m + e_m$ with a policy-specific bias b_m, a bias common to j and m cancels in $\hat{D}$ while a bias that differs between the two policies—for instance one concentrated on documents that only policy j retrieves—lowers ρ regardless of the judge's overall accuracy.

Empirical test. We measured ρ and the realised gain (cross-fitted λ, $T = 10$ audited queries, 300 random audits) on 54 (collection, pair, judge) settings: four BEIR development pools under a legacy and a modern retrieval stack, five fully judged pools (TREC DL 2019, 2020; TREC-COVID; Touché 2020; DBpedia-Entity), and two judges (Qwen3-Reranker-0.6B and Qwen3-8B with the UMBRELA prompt). Figure 1 shows the result: $\mathrm{corr}(\text{gain}, \rho) = 0.78$ and $\mathrm{corr}(\text{gain}, \text{pair accuracy}) = -0.02$. The relation is not an artefact of pooling heterogeneous groups: demeaned within the 14 (collection, judge) groups it is 0.66 and positive in all 14; and for the 13 (collection, pair) settings in which switching from the reranker to the LLM judge raised ρ, the gain rose in all 13.

When does the gain materialise? Proposition 1 is asymptotic in N. On DBpedia-Entity (399 queries) and TREC DL 2021–23 (211 queries) we manipulated three quantities: the judge's ρ (by replacing a fraction

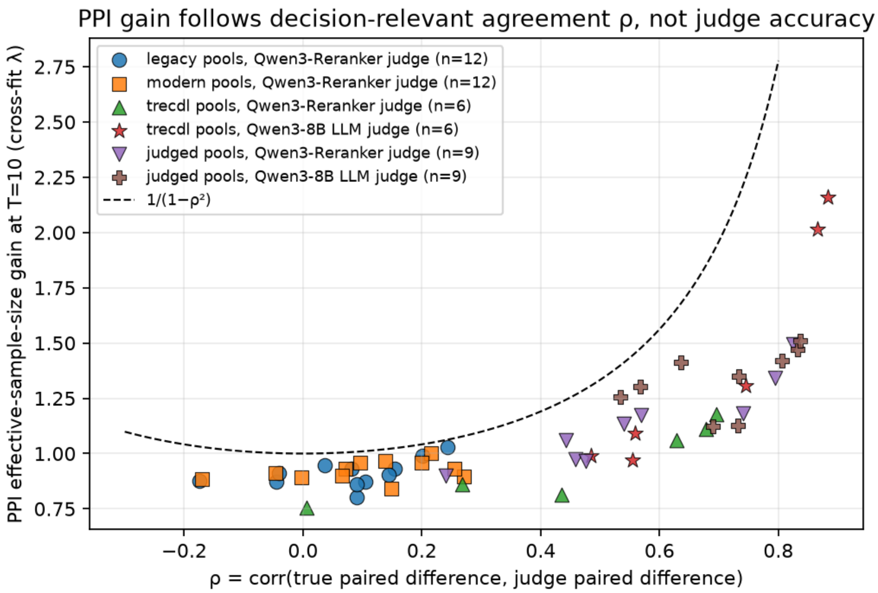

Figure 1: Realised PPI effective-sample-size gain at $T = 10$ against ρ, 54 settings, cross-fitted λ. The dashed curve is $1/(1 - \rho^2)$. Colour/marker distinguish pools and judges. Gains track ρ (0.78) and not accuracy (−0.02).

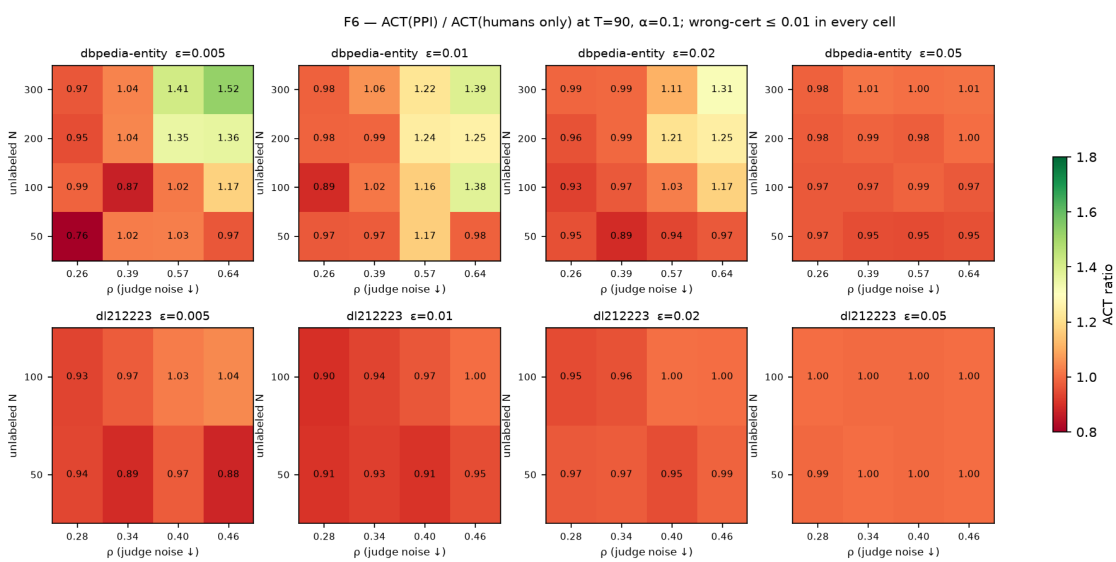

Figure 2: Mechanism map: ACT(PPI)/ACT(humans) at $T = 90$ over ρ (judge-label noise), unlabeled pool size and ε; wrong-certificate rate ≤ 0.01 in every cell. Gains need $\rho \geq 0.55$, $N_{\text{unlab}} \geq 2T$ and a hard decision simultaneously.

of its labels with base-rate coin flips), the unlabeled size N_{unlab}, and ε, and measured the ACT ratio of the PPI certificate to the human-only certificate at $T = 90$ (Figure 2). The ratio exceeds 1.2 only when all of $\rho \geq 0.55$, $N_{\text{unlab}} \geq 200 \approx 2T$, and $\varepsilon \leq 0.02$ (a decision the human-only certificate cannot settle easily) hold; otherwise it is 0.9–1.05. Every cell has wrong-certificate rate ≤ 0.01. On TREC DL 2021–23 the judge reaches $\rho \leq 0.46$ and $N_{\text{unlab}} \leq 100$, so no cell shows a gain—which is exactly what the pre-registered evaluation found (Section 7).

Table 1: Self-preference test at fixed policies: mean over 15 policy pairs per judge, split by whether the pair involves the reranker-defined policy. "Bias" is the mean of predicted minus true paired difference.

Judge	Pairs	ρ	in-band / out-of-band error	bias	PPI gain
Qwen3-Reranker (identical model)	without / with	0.57 / 0.25	0.36 / 0.32 → 0.41 / 0.31	−0.03 / **−0.14**	1.07 / 0.95
Qwen3-8B (same family)	without / with	0.69 / 0.48	0.28 / 0.24 → 0.29 / 0.23	0.00 / −0.04	1.19 / 1.04
Mistral-7B (other family)	without / with	0.38 / 0.30	0.34 / 0.34 → 0.37 / 0.33	+0.02 / −0.04	1.10 / 1.06
MPNet rank rule (other family)	without / with	0.39 / 0.21	0.39 / 0.35 → 0.40 / 0.35	−0.05 / +0.03	1.03 / 0.94

3.1 Judges built from the model that defines a policy

We added to the menu a fourth policy, `rr_thresh`, that returns every pooled document whose Qwen3-Reranker score exceeds a threshold tuned on the audit, so that one policy is defined by a model in the judges' family. With policies fixed at their population parameters, we compared five judges on the six policy pairs of five fully judged pools (Table 1). Two signatures are specific to the *identical-model* judge: it over-credits its own policy by −0.14 on the paired F1 difference (all other judges −0.04 to +0.03), and its error rate inside the documents on which the two policies disagree rises from 0.36 to 0.41. The drop in ρ for pairs involving `rr_thresh` appears for every judge, including the different-family MPNet rule, so it is largely a property of that policy's paired differences rather than of self-preference. Validity is unaffected: with the four-policy menu the split certificate produced 0/500 wrong certificates for both the identical-model and the same-family judge, and the identical-model judge yielded no efficiency gain (ACT 246 vs 245 human-only) while the 8B judge did (328).

4 A certificate that is valid for any judge

Why the earlier certificate was not. The certificate used in our earlier development (leave-one-out cross-fitted utilities, paired bootstrap, Bonferroni over looks $\times(|\mathcal{M}|-1)$ comparisons) chose the candidate on the same data used for the bound. With exact t bounds on synthetic data this alone raises the per-look miss rate from the 0.020 budget to 0.049; correcting over all $|\mathcal{M}|(|\mathcal{M}|-1)$ ordered pairs restores 0.014. On real data at 500 repeats the earlier certificate's wrong-certificate rate was 0.146 $[0.116, 0.180]$ on one development pool and 0.180 $[0.147, 0.217]$ on another, both above $\alpha = 0.10$ with the whole interval.

Split certificate. At look t the audited queries are split into a training half A_t and a validation half B_t. Every data-driven choice—policy parameters, the candidate $\hat{m}_t$ (largest lower bound on A_t), and whether to adopt the judge—is made on A_t; the bound is computed on B_t only, as a one-sided t bound on the paired differences (humans only) or as a PPI++ bound with λ cross-fitted inside B_t and the unlabeled mean taken outside B_t (with the judge).

Proposition 2 (Conditional validity for any judge)**.** *Conditionally on the σ-field generated by A_t and the decisions taken on it, and for every fixed judge f, each $U_{t,j}$ is a level-$(1-\alpha')$ upper confidence bound for $\mu_j - \mu_{\hat{m}_t}$ up to the t-approximation; hence* $\Pr(\exists t:\ \mathrm{ACT}_t \wedge \mu_{\text{best}} - \mu_{\hat{m}_t} > \varepsilon) \le \alpha$ *regardless of f, of the adoption rule, and of whether the ρ-based adoption score covers its target.*

The proof (Appendix B) uses only that B_t is independent of the choices made on A_t, that a cross-fitted λ is independent of the fold it is applied to, and a union bound over all ordered pairs and looks. The adoption score never enters the bound; a bad adoption decision changes which valid bound is used, hence only efficiency.

Empirical validity. Across 24 (collection, judge, utility) development cells at 500 repeats, the split certificate's wrong-certificate rate is $\le 4/500 = 0.008$ (upper 95% limit ≤ 0.020). With an *adversarial* judge (the reranker's probabilities inverted, pair accuracy 0.34) it is 2/500 and the PPI certificate degenerates to the human-only one (156 vs 158 cumulative ACT at $T = 190$) because $\lambda \to 0$. Figure 3 shows the cumulative ACT curves by judge on DBpedia-Entity.

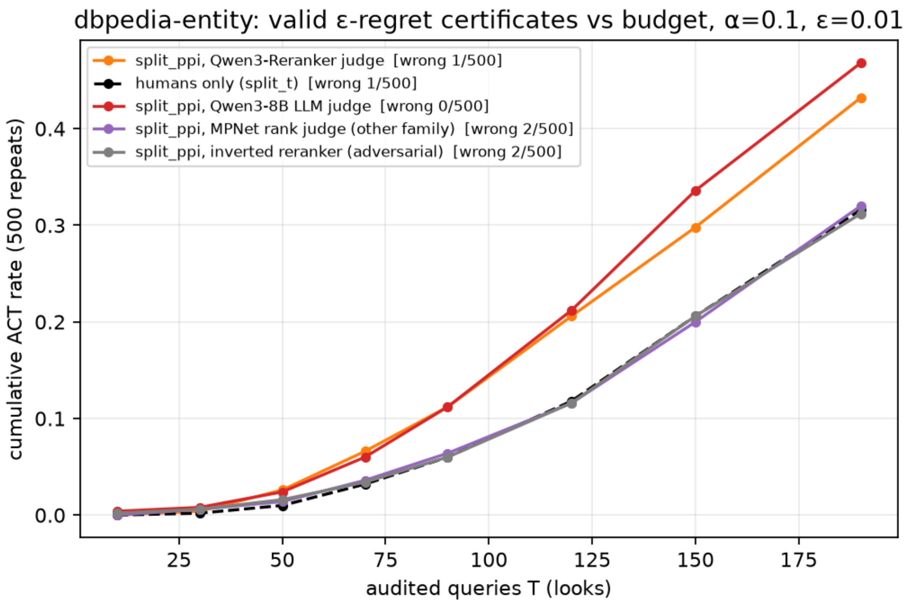

Figure 3: Cumulative ACT rate of the split certificate against the audit budget on DBpedia-Entity (500 repeats, $\varepsilon = 0.01$). Judges change the slope, not the validity: wrong certificates $\leq 2/500$ for every curve, including the inverted (adversarial) judge.

Always using the judge can cost efficiency; a pilot rule protects it. PPI++ with λ estimated on small folds is not guaranteed to be at least as efficient as the human-only bound when the unlabeled pool is small relative to the audit. On TREC DL 2021–23 with the reranker judge (pair accuracy 0.43 there), the always-PPI certificate reached 256/500 cumulative ACT at $T = 90$ against 313/500 for humans only; using the finite-N optimal λ recovered 282, and a variance guard 266, but neither closed the gap. A rule that adopts the judge only when a BCa lower bound on ρ from the training half (at least 30 queries) implies a gain above 1.1 adopted the reranker in 1.3% of runs and stayed at 309. The rule's value is therefore efficiency protection and the avoidance of judge inference cost; validity does not depend on it. The BCa bound covers its target at 0.88–0.92 for pilots of 30 queries or more (Appendix C); we call it an adoption *score*, not a guarantee.

5 Where to spend human judgments

Proposition 3 (Decision weights of documents)**.** *For prefix policies a, b with cutoffs $k_a \leq k_b$ on the same ranked pool and a utility $u_m = \sum_{d \in R_m} y_d / Z_m$ whose normaliser does not depend on the labels, $u_a - u_b = \sum_d w(d)\, y_d$ with $w(d) = 1/Z_a - 1/Z_b$ on common documents, $w(d) = -1/Z_b$ on the band $R_b \setminus R_a$, and 0 elsewhere. Exact band sufficiency holds iff $Z_a = Z_b$; otherwise the common documents carry weight $|1/Z_a - 1/Z_b|$, small when the cutoffs are close and never zero. For set-F1 the normaliser depends on the labels through n_G and the relation holds only after linearisation (Appendix A).*

An earlier draft claimed band sufficiency for precision; a reviewer's counterexample ($R_A = \{a\}$, $R_B = \{a, b\}$, b irrelevant: the difference is 0.5 or 0 depending on a) showed this is false, and a band-only audit built on that claim yielded no gain at equal budget. Proposition 3 is the corrected statement.

Estimators. Let π_d be the inclusion probability of a human label on document d and $\hat{\jmath}_d$ the judge's label. The Horvitz–Thompson estimator $\hat{D}_{\mathrm{HT}} = \sum_{d \text{ sampled}} w(d) y_d / \pi_d$ and the control-variate estimator $\hat{D}_{\mathrm{CV}} = \sum_d w(d) \hat{\jmath}_d + \sum_{d \text{ sampled}} w(d)(y_d - \hat{\jmath}_d)/\pi_d$ are both unbiased for $D(q)$ for any judge; their variances are $\sum_d w(d)^2 y_d^2 (1-\pi_d)/\pi_d$ and $\sum_d w(d)^2 (y_d - \hat{\jmath}_d)^2 (1-\pi_d)/\pi_d$. The judge helps exactly on the documents where it

Table 2: J_{50}: human-judged pairs to reach 50% ACT at equal error control (precision at cutoff, $\varepsilon = 0.02$, $\alpha = 0.10$, 300 draws; wrong-certificate rate ≤ 0.007 in every cell). "n.r." = not reached within the largest budget run. Judge: Qwen3-8B unless stated.

Method	DBpedia-Entity	DBpedia-Entity, reranker judge	TREC DL 21–23
Uniform document sampling (humans)	n.r. (> 2,570)	n.r.	3,207
Decision-weight importance sampling (humans)	1,990	1,990	1,273
Stratified by pilot variance (humans)	2,144	2,144	1,388
Active inference, calibrated-judge rule + CV	2,020	2,054	1,562
Active inference, residual-estimated rule + CV	1,610	1,715	1,340
Active inference, robust mixture + CV	1,556	1,628	1,311
Decision-weight sampling + judge control variate	**1,569**	**1,705**	**1,256**

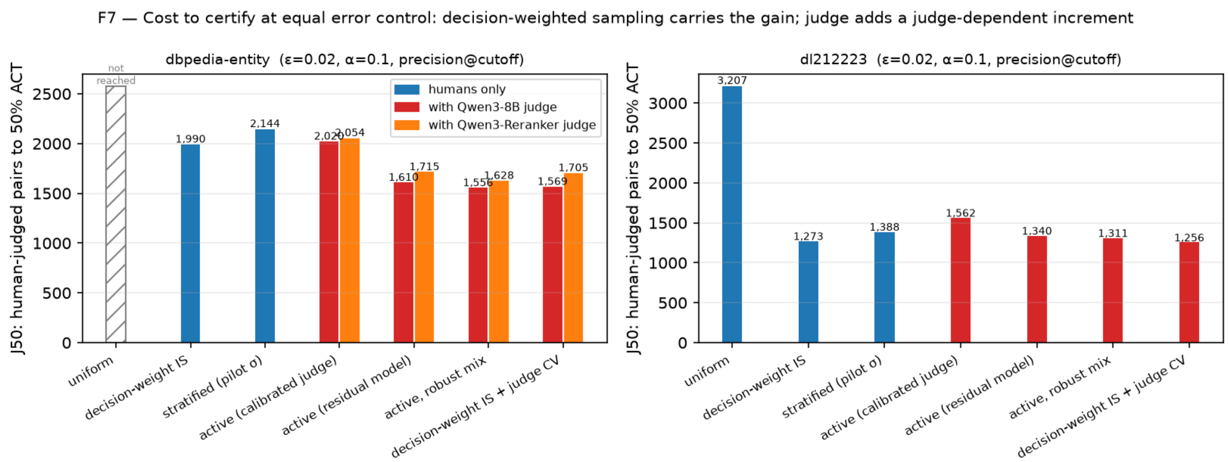

Figure 4: J_{50} by method (precision at cutoff, $\varepsilon = 0.02$). The bulk of the saving comes from decision-weight importance sampling; the judge adds a judge-dependent increment as a control variate; the calibrated-judge active rule is the worst judge-assisted arm.

is right, weighted by decision weight, and never affects unbiasedness. Query-level certificates (Proposition 2) then apply with $\hat{D}$ in place of a fully audited D.

5.1 Relation to active statistical inference

$\hat{D}_{\mathrm{CV}}$ is the active (prediction-powered) estimator of Zrnic & Candès (2024) written for the linear functional $\sum_d w(d)y_d$. Under a budget $\sum_d \pi_d = b$ its variance-minimising allocation is $\pi_d^\star \propto |w(d)|\sqrt{\mathbb{E}[(y_d - \hat{\jmath}_d)^2]}$. Our rule $\pi_d \propto |w(d)|$ is the special case of a constant residual model; the "calibrated-judge" rule uses $\hat{\jmath}_d(1-\hat{\jmath}_d)$; the "residual-estimated" rule fits $\mathbb{E}[(y-\hat{\jmath})^2]$ on the pilot; Li et al. (2025) mix an uncertainty-based rule with a safe distribution, which here is $\pi_d \propto |w(d)|$ itself. What is specific to this paper is not the estimator but the identification of the functional (the decision weights) and the certificate that consumes it.

5.2 Comparison at equal human budget

All arms below share a 20-query fully labelled pilot (counted in the cost) that fixes policy parameters, the candidate, and any residual model; the same expected number of human-labelled documents; and the same certificate. Table 2 reports J_{50} for precision at cutoff, $\varepsilon = 0.02$, and Figure 4 shows it graphically.

Three readings follow. First, the large saving is obtained without any judge: decision-weight importance sampling lowers the per-query estimator variance 2.9–4.4× relative to uniform document sampling and halves J_{50} or more; a pilot-estimated stratification adds nothing beyond $|w|$. Second, the judge adds a judge-dependent increment only as a control variate: −21% with Qwen3-8B, −14% with the reranker, none on TREC DL 21–23 where $\rho \approx 0.46$. Unbiasedness of $\hat{D}_{\mathrm{CV}}$ does not by itself guarantee that a poor judge cannot add variance; the second pre-registered evaluation found a 16% loss with an adversarial judge, which a pilot-estimated coefficient on the control variate removes (Section 7). Third, the faithful active-inference rule (residual model from the pilot) and its robust mixture match decision-weight sampling within noise (variance ratios 0.95–1.01), whereas the calibrated-judge rule is 1.3–1.5× worse because the judge is confidently wrong on some decisive documents—the failure mode anticipated by Li et al. (2025). We also asked whether allocating labels adaptively to the comparisons that still bind the decision helps when the menu has four policies: it did not (static shared allocation 1,525, adaptive 1,463, oracle allocation 1,245 on DBpedia-Entity), so we report it as a negative result (Section 8).

Validity stress tests. Because a wrong-certificate rate of zero is uninformative when the candidate's true regret is below ε, we ran two stress designs for every document-level arm: forcing the candidate to the true worst policy (type-I error ≤ 0.007; an easy test since the gap is then far above ε) and a *boundary* design that sets the candidate to the runner-up and ε to 0.9× its true regret, so that every ACT is a type-I error (realised ε 0.015–0.034): type-I error ≤ 0.03 for every arm on both collections, below α by the Bonferroni margin.

6 Experimental protocol

Pools and policies. Policies are cutoff rules over a shared pool: an expected-F1 gate with a target-calibrated scalar, a global score threshold, a learned truncation, and (for the self-preference test) a reranker-score threshold. Pools are built by the legacy protocol of the base study (BM25 plus three sentence-transformer retrievers, top-30 each, union) and a modern variant with Qwen3-Embedding-0.6B in place of the weakest retriever; the relevance model is a calibrated logistic regression on four pool features trained on four BEIR development collections (nfcorpus, scifact, arguana, cqadupstack-android) and never on a target. *Fully judged* pools take the NIST/BEIR-judged documents of each query as the corpus, so every pooled pair carries a human label: TREC DL 2019 (43 queries) and 2020 (54), TREC-COVID (50), Touché 2020 (49), DBpedia-Entity (399), and TREC DL 2021–2023 on MS MARCO v2 (211). The BEIR development pools have sparse judgments; results on them are used only for the ρ analysis and are marked as such.

Judges. Qwen3-Reranker-0.6B (probability of "yes"), Qwen3-8B and Mistral-7B-Instruct-v0.3 with the UMBRELA 0–3 prompt (Upadhyay et al., 2024) read from the next-token distribution over the digit tokens, an MPNet cosine rank rule (label-free, different family), and the inverted reranker as an adversarial control. Batched inference with left padding requires explicit position ids; without them one collection's judgments collapsed, which we detected and re-ran (Appendix C).

Repeats and randomness. 500 repeats for sequential certificates and 300 draws for fixed-budget designs; every method has its own random stream so adding a method never perturbs another's numbers; all seeds are recorded.

7 Pre-registered external evaluations

Table 3 summarises the three locks; the subsections give the details, and Table 4 reports the confirmatory J_{50} values. Before downloading TREC DL 2021–2023, we committed a lock fixing the target, pooling rule, training data, $\varepsilon = 0.01$, $\alpha = 0.10$, the look grid, the three methods (humans only, always-PPI, rule-based adoption), the judge, 500 repeats, and three criteria: validity (Clopper–Pearson upper limit of the wrong-certificate rate ≤ 0.15 and point estimate $\leq \alpha$ for all three methods), efficiency (always-PPI $\geq 1.2\times$ humans-only in cumulative ACT at the largest look; rule-based $\geq$ humans-only), and adversarial rejection (the rule adopts the inverted judge in $\leq 5\%$ of runs). The pool has 211 queries, so looks are capped at $T = 90$.

Validity passed for every method and judge (0/500; upper limit 0.007), and the adversarial-rejection criterion passed (0%). The efficiency criteria did not: always-PPI reached 318 against 313 (1.02×), and the rule-based

Table 3: The three pre-registered evaluations. Every criterion was fixed in a lock committed before the target data were analysed; outcomes are reported as written.

Lock / target	What was tested	Validity	Efficiency criteria
v0.4 / TREC DL 2021–23 (211 q)	query-level certificates: humans, always-PPI, rule-based; judges 8B, reranker, inverted	**pass**: 0/500 wrong for all methods and judges; rule adopted the inverted judge in 0%	**not met**: always-PPI 318 vs 313 (1.02×, needed 1.2×); rule 308. Cause: $\rho \leq 0.5$, unlabeled pool 121, easy gaps (Fig. 2)
v0.5 / TREC CAsT 2019 (159 turns)	document-level: weighted sampling, judge CV, active inference	**pass**: boundary type-I ≤ 0.013; judge CV never worse (max-ACT rule)	**not evaluable**: no arm reached 50% ACT within the locked grid (best 0.43); ordering replicated; post-hoc grid: J_{50} 2,750 vs uniform n.r.
v0.6 / ANTIQUE (200 q)	λ-corrected CV; always-defined criteria	**pass**: boundary type-I ≤ 0.010; λ-CV never worse for 3 judges; defect fix confirmed; active rule = CV	**not met** on the max-budget criterion (saturated: 0.96 vs 0.92); co-registered J_{50} 776 vs 1,555 (pass); judge -26%

Table 4: Confirmatory collections, document-level precision at cutoff, $\varepsilon = 0.02$: J_{50} (human-judged pairs to 50% ACT) by judge. CAsT rows use the locked grid plus the post-hoc budgets 120/150; ANTIQUE rows are fully pre-registered. "n.r." = not reached; wrong-certificate rate ≤ 0.003 in every cell.

Collection	Method	humans	Qwen3-8B	Reranker	inverted
TREC CAsT 2019	Uniform document sampling (humans)	n.r.	n.r.	n.r.	n.r.
	Decision-weight importance sampling (humans)	2,784	n.r.	n.r.	n.r.
	Stratified by pilot variance (humans)	2,947	n.r.	n.r.	n.r.
	Active inference, calibrated-judge rule + CV	n.r.	2,731	n.r.	n.r.
	Active inference, residual-estimated rule + CV	n.r.	2,807	n.r.	n.r.
	Active inference, robust mixture + CV	n.r.	2,779	n.r.	n.r.
	Decision-weight sampling + judge CV (coefficient 1)	n.r.	2,694	2,924	3,275
	Decision-weight sampling + judge CV (pilot λ)	n.r.	2,610	2,686	2,719
ANTIQUE	Uniform document sampling (humans)	1,555	n.r.	n.r.	n.r.
	Decision-weight importance sampling (humans)	776	n.r.	n.r.	n.r.
	Stratified by pilot variance (humans)	747	n.r.	n.r.	n.r.
	Active inference, calibrated-judge rule + CV	n.r.	738	n.r.	n.r.
	Active inference, residual-estimated rule + CV	n.r.	598	n.r.	n.r.
	Active inference, robust mixture + CV	n.r.	602	n.r.	n.r.
	Decision-weight sampling + judge CV (coefficient 1)	n.r.	581	687	826
	Decision-weight sampling + judge CV (pilot λ)	n.r.	576	627	733

method 308. The mechanism map explains the outcome: on this collection the judge's ρ is 0.40–0.52, the unlabeled pool after auditing is 121 queries, and the policy gaps are large enough that humans alone certify 63% by $T = 90$—outside the region where Figure 2 predicts a gain. We report the result as pre-registered; the document-level methods of Section 5 were developed after this evaluation and have not yet been tested on data unseen during their development.

7.1 Second pre-registered evaluation: document-level methods on TREC CAsT 2019

After the document-level methods of Section 5 were developed, a second lock fixed a target that had never been used: TREC CAsT 2019 evaluation turns restricted to the judged MS MARCO v1 passages (159 turns with a grade-≥ 2 passage, 48.7 pooled documents per turn, resolved utterances as queries), the same pooling rule and training data, the Qwen3-8B judge, a budget grid of $\{30, 45, 60, 90\}$ full-query equivalents, $\varepsilon \in \{0.01, 0.02\}$, 300 draws, and four criteria: (P1) boundary type-I error $\leq \alpha$ for every arm, (P2)

Table 5: Pre-registered evaluation on TREC DL 2021–2023 (500 repeats, looks ≤ 90, $\varepsilon = 0.01$). Cumulative ACT out of 500 and wrong certificates.

Judge	Method	ACT by T=50 / 70 / 90	wrong / 500	adoption
—	humans only	84 / 198 / 313	0	—
Qwen3-8B	always-PPI	105 / 220 / 318	0	—
Qwen3-8B	rule-based	84 / 203 / 308	0	26%
Qwen3-Reranker	always-PPI	52 / 136 / 256	0	—
Qwen3-Reranker	rule-based	84 / 194 / 309	0	1.3%
inverted	always-PPI	70 / 182 / 304	0	—
inverted	rule-based	84 / 198 / 313	0	0%
earlier in-sample certificate	—	434 / 494 / 498	7	—

J_{50}(weighted) $\leq 0.6\, J_{50}$(uniform), (P3) the judge control variate never worse than weighted sampling for three judges (8B, reranker, inverted), and (P4) faithful active inference within 10% of the weighted control variate. The judge reached pair accuracy 0.79 and $\rho = 0.62$, the most favourable judge conditions in this paper.

P1 and P3 passed (maximum type-I error 0.013; judge arms within 0.05 of weighted sampling in ACT at the largest budget). P2 and P4 could not be evaluated as written, because *no* arm reached 50% ACT within the locked budget grid (best 0.43 at $\varepsilon = 0.02$): conversational queries make the policy gaps small, and the grid had been sized on easier collections. By the lock's own rule this counts as not met, and it is a lesson about pre-registration rather than about the methods—criteria must include a quantity that is always defined, such as ACT at the largest budget. Within the locked grid the ordering of Table 2 replicated exactly: uniform sampling 0.09, decision-weight sampling 0.39, stratified 0.37, calibrated-judge active rule 0.22 (with control variate 0.39), residual-estimated active rule 0.43, decision-weight sampling with the 8B control variate 0.42. A post-hoc extension of the grid to 150 full-query equivalents (reported as post hoc) gives J_{50} of 2,750 judged pairs for decision-weight sampling against *not reached* for uniform sampling, 2,632 with the 8B control variate (-4%, the small increment expected at $\rho = 0.62$), and 2,808 for the residual-estimated active rule. The query-level methods, run for continuity, showed the same picture as Figure 2 predicts for a hard decision with $\rho = 0.62$: cumulative ACT at $T = 70$ of 48/500 for humans only against 100/500 with the 8B judge, 0/500 wrong certificates.

The post-hoc extension also exposed a defect: with the inverted judge the coefficient-1 control variate needed 16% more judgments than weighted sampling alone (3,194 vs 2,750). Unbiasedness does not protect efficiency when the judge is adversarial, exactly as PPI without λ would not. We therefore estimate a coefficient $\lambda \in [0, 1]$ on the pilot (regression of true on judge paired differences) and use $\lambda\hat{\jmath}$ as the control variate, so that $\lambda \to 0$ recovers Horvitz–Thompson sampling. This change was made after the lock and its evaluation is post hoc: with the inverted judge the coefficient-λ control variate needs 2,720 judgments against 2,948 for weighted sampling in the same run (coefficient-1: 3,362), and on DBpedia-Entity 2,024 against 2,094 (coefficient-1: 2,388); with the 8B judge it matches or slightly improves on the coefficient-1 version (2,611 vs 2,625; 1,586 vs 1,628). It is the first item of a third lock.

7.2 Third pre-registered evaluation: the λ-corrected control variate on ANTIQUE

The third lock targeted ANTIQUE (Hashemi et al., 2020), a non-factoid question-answering test set never used in development (200 questions, 31.6 pooled answers per question, relevant = grade ≥ 3 of 4), with a wider budget grid $\{30, 60, 90, 120, 150\}$, criteria defined on ACT at the largest budget so that they are always evaluable, and the same judges. The Qwen3-8B judge, prompted on a 0–3 scale that does not match ANTIQUE's 1–4 scale, reached a pair accuracy of only 0.47 at the fixed threshold—yet $\rho = 0.59$.

Validity passed (P1: boundary type-I error ≤ 0.010). The λ-corrected control variate was never worse than weighted sampling for any of the three judges (P3), it was never worse than the coefficient-1 variate with the inverted judge (P4), and the residual-estimated active rule matched it (P5: 0.94 vs 0.94). The always-defined efficiency criterion (P2: ACT(weighted) $\geq 1.5 \times$ ACT(uniform) at the largest budget) failed for the opposite reason to CAsT: the grid was large enough that every arm saturated (0.96 vs 0.92), whereas the co-registered J_{50} read 776 against 1,555 judged pairs (0.50). With the judge, J_{50} fell to 576 (-26%; reranker 627; inverted 733), and at the smallest budget the judge raised ACT from 0.28 to 0.50. The coefficient-1 defect reappeared at the 60-equivalent budget with the inverted judge (0.867 vs 0.913 for weighted sampling) and the λ correction removed it (0.930). The query-level certificate, run for continuity, gave 146/500 against 111/500 cumulative ACT at $T = 90$ with 5–6/500 wrong certificates. The lesson for pre-registration is symmetric to CAsT's: an always-defined criterion must be evaluated at a budget where the human-only arm is in its informative range, which we now fix by rule.

8 Negative and bounded results

Four things did not work, and we think they are worth knowing. (1) A band-only hybrid predictor (human labels on the disagreement band, judge elsewhere) raises ρ from 0.6 to 0.9 for set-F1 but gives no gain at equal budget, because the rectifier still needs fully labelled queries; for precision it does not raise ρ at all, as Proposition 3 implies. (2) Adaptive allocation of document labels across the comparisons of a four-policy menu equals static shared allocation, and even an oracle allocation gains only 5–18%, because dominated comparisons are settled by few labels. (3) Two attempts to make always-PPI safe by adjusting λ (finite-N optimum, variance guard) reduced but did not remove its loss with a poor judge; the pilot adoption rule did. (4) An earlier version of this paper reported that judge-uncertainty sampling "harms"; a faithful implementation with a pilot-estimated residual model does not, and we retract the general claim—only the calibrated-judge rule harms.

9 Related work

Prediction-powered and active inference. PPI (Angelopoulos et al., 2023a) and PPI++ (Angelopoulos et al., 2023b) correct AI predictions with human labels and tune λ to bound the loss from poor predictions; active statistical inference (Zrnic & Candès, 2024) chooses which points to label while keeping the guarantee; Li et al. (2025) mix uncertainty-based sampling with a safe distribution; Sfyraki & Wang (2026) find constant-probability sampling competitive for sequential mean estimation; Kilian et al. (2025) give anytime-valid PPI. We use this machinery unchanged. What we add is the functional the certification problem requires (closed-form document weights from a policy pair), a measurement of how the calibrated-judge rule fails on decisive documents, and the observation that a judge used only as a control variate leaves the certificate valid for any judge.

AI judges for IR evaluation. Oosterhuis et al. (2024) build PPI and conformal intervals for IR metrics from LLM judgments; Chatzi et al. (2024) rank LLMs from LLM comparisons with human anchoring; Gligorić et al. (2024) use LLM confidence to allocate human annotation; Balog et al. (2025) document that LLM judges favour LLM-based rankers; UMBRELA (Upadhyay et al., 2024) is the source of our prompt; Durmazkeser et al. (2026) select among many LLMs with few labels. These treat the judge as a fixed predictor of a metric or ranking. We add that for deployment decisions the judge's value is set by ρ of paired differences, not accuracy; that this varies strongly by judge family and collection (Mistral-7B collapses on Touché while matching Qwen3-8B on DBpedia); and the measured self-preference of the identical-model judge.

Sequential testing and best-arm identification. Adaptive learn-then-test (Zecchin & Simeone, 2024), fixed-confidence BAI (Jamieson & Nowak, 2014; Kaufmann et al., 2016), and ε-best-answer identification (Jourdan & Degenne, 2022) give stopping rules; multi-task PPI (Emmenegger et al., 2026) shares a predictor across tasks. We use a plain union bound over looks and ordered pairs and, in return, handle the winner's curse of data-driven candidate choice explicitly; the label-sharing structure across comparisons (one document label with different weights per comparison) is outside these frameworks, and our bounded exploration of it was negative.

IR evaluation cost. Topic-set size design (Sakai, 2016), few-topic prediction (Guiver et al., 2009), active judgment sampling (Li & Kanoulas, 2017), active comparison of two models (Sawade et al., 2012), and active testing (Kossen et al., 2021) all trade labels for evaluation precision. We inherit Sawade et al.'s principle of labelling where the difference lives, carry it to the document level with a certification target, and account for costs by audit unit and utility.

10 Limitations and conclusion

Our pools consist of judged documents only, so conclusions concern deployment decisions over judged candidate sets; sparse-judgment collections are used only for the ρ analysis. Precision at cutoff is the primary utility for the document-level results; set-F1 is treated by a bias-corrected linearisation whose remainder we measure (up to 3ε on one collection). The certificate uses Bonferroni over looks rather than an anytime-valid sequence. The document-level methods have not yet been evaluated on data unseen during their development; a second pre-registered evaluation is the obvious next step. Two judge families were used. Within these limits, the paper's message is a single one: define the estimand that certification actually requires, and the existing tools—importance sampling by decision weight, a judge as control variate, a split certificate—deliver most of what an AI judge can offer, with validity that does not depend on the judge and with an honest account of when the judge does not help.

References

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023a.

Anastasios N. Angelopoulos, John C. Duchi, and Tijana Zrnic. PPI++: Efficient prediction-powered inference. *arXiv preprint arXiv:2311.01453*, 2023b.

Krisztian Balog et al. Rankers, judges, and assistants: Towards understanding the interplay of LLMs in information retrieval evaluation. *arXiv preprint arXiv:2503.19092*, 2025.

Ivi Chatzi, Eleni Straitouri, Suhas Thejaswi, and Manuel Gomez-Rodriguez. Prediction-powered ranking of large language models. In *Advances in Neural Information Processing Systems*, 2024.

Yavuz Durmazkeser, Patrik Okanovic, Andreas Kirsch, Torsten Hoefler, and Nezihe Merve Gürel. Large language model selection with limited annotations. *arXiv preprint arXiv:2605.24981*, 2026.

Nicolas Emmenegger, Ellery Stahler, and Chara Podimata. Prediction-powered inference across many tasks for AI evaluation & social science research. *arXiv preprint arXiv:2605.29249*, 2026.

Kristina Gligorić, Tijana Zrnic, Cinoo Lee, Emmanuel J. Candès, and Dan Jurafsky. Can unconfident LLM annotations be used for confident conclusions? *arXiv preprint arXiv:2408.15204*, 2024.

John Guiver, Stefano Mizzaro, and Stephen Robertson. A few good topics: Experiments in topic set reduction for retrieval evaluation. *ACM Transactions on Information Systems*, 27(4), 2009.

Helia Hashemi, Mohammad Aliannejadi, Hamed Zamani, and W. Bruce Croft. ANTIQUE: A non-factoid question answering benchmark. In *European Conference on Information Retrieval*, 2020.

Kevin Jamieson and Robert Nowak. Best-arm identification algorithms for multi-armed bandits in the fixed confidence setting. In *48th Annual Conference on Information Sciences and Systems (CISS)*, 2014.

Marc Jourdan and Rémy Degenne. Choosing answers in ε-best-answer identification for linear bandits. In *International Conference on Machine Learning*, 2022.

Emilie Kaufmann, Olivier Cappé, and Aurélien Garivier. On the complexity of best-arm identification in multi-armed bandit models. *Journal of Machine Learning Research*, 17(1):1–42, 2016.

Valentin Kilian et al. Anytime-valid, Bayes-assisted, prediction-powered inference. In *Advances in Neural Information Processing Systems*, 2025.

Jannik Kossen, Sebastian Farquhar, Yarin Gal, and Tom Rainforth. Active testing: Sample-efficient model evaluation. In *International Conference on Machine Learning*, 2021.

Dan Li and Evangelos Kanoulas. Active sampling for large-scale information retrieval evaluation. In *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management*, 2017.

Puheng Li et al. Robust sampling for active statistical inference. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2511.08991.

Harrie Oosterhuis, Rolf Jagerman, Zhen Qin, Xuanhui Wang, and Michael Bendersky. Reliable confidence intervals for information retrieval evaluation using generative A.I. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024.

Tetsuya Sakai. Topic set size design. *Information Retrieval Journal*, 19(3):256–283, 2016.

Christoph Sawade, Niels Landwehr, and Tobias Scheffer. Active comparison of prediction models. In *Advances in Neural Information Processing Systems*, 2012.

Maria-Eleni Sfyraki and Jun-Kun Wang. Revisiting active sequential prediction-powered mean estimation. In *International Conference on Learning Representations*, 2026.

Shivani Upadhyay, Ronak Pradeep, Nandan Thakur, Nick Craswell, and Jimmy Lin. UMBRELA: UMbrela is the (open-source reproduction of the) Bing RELevance Assessor. *arXiv preprint arXiv:2406.06519*, 2024.

Matteo Zecchin and Osvaldo Simeone. Adaptive learn-then-test: Statistically valid and efficient hyperparameter selection. *arXiv preprint arXiv:2409.15844*, 2024.

Tijana Zrnic and Emmanuel J. Candès. Active statistical inference. *Proceedings of the National Academy of Sciences*, 2024. arXiv:2403.03208.

A Set-F1: linearised document-level auditing

For set-F1, $D(y) = 2\,\mathrm{tp}_a/(k_a+n_G) - 2\,\mathrm{tp}_b/(k_b+n_G)$ is a ratio of label sums. We linearise at the judge labels: $w_d = \partial D/\partial y_d|_{y=\hat{y}} = 2\cdot\mathbf{1}[d\in R_a]/(k_a+\hat{n}_G) - 2\cdot\mathbf{1}[d\in R_b]/(k_b+\hat{n}_G) - 2\hat{\mathrm{tp}}_a/(k_a+\hat{n}_G)^2 + 2\hat{\mathrm{tp}}_b/(k_b+\hat{n}_G)^2$, so every pooled document has a non-zero weight through n_G; $\hat{D}_{\mathrm{lin}} = D(\hat{y}) + \sum_{d \text{ sampled}} w_d(y_d-\hat{y}_d)/\pi_d$ with $\pi_d \propto |w_d|$. The remainder $D - \hat{D}_{\mathrm{lin}}$ is measured on the fully labelled pilot and its mean is added as a bias correction with its standard error folded into the bound. A humans-only plug-in alternative estimates $\mathrm{tp}_a, \mathrm{tp}_b, n_G$ by Horvitz–Thompson and plugs them into F1.

Measured remainders: DBpedia-Entity $+0.002$ (mean absolute 0.011), DBpedia with the reranker judge -0.012 (0.018), TREC DL 21–23 -0.029 (0.030), i.e. three times $\varepsilon = 0.01$ on the last collection, where the uncorrected linearised certificate is not trustworthy. At equal document budget (60 full-query equivalents, $\varepsilon = 0.01$, 300 draws) the ACT rates on DBpedia-Entity were: full-query audit 0.23, plug-in HT (humans) 0.78, query-level PPI 0.33, linearised CV 0.55, bias-corrected linearised CV 0.32; on TREC DL 21–23 (30 full-query equivalents): 0.10, 0.72, 0.10, 0.86 (inflated by the remainder), 0.42. The boundary stress test of Section 5 gave type-I error ≤ 0.013 for all set-F1 arms at realised $\varepsilon \approx 0.03$; it does not probe $\varepsilon = 0.01$ on TREC DL 21–23, where only the bias-corrected variant should be used.

B Proofs

Proposition 1. The two sample means are independent; $\mathrm{Var}[\overline{(D-\lambda\hat{D})}_n] = (\sigma^2 - 2\lambda\rho\sigma\hat{\sigma} + \lambda^2\hat{\sigma}^2)/n$ and $\mathrm{Var}[\lambda\overline{\hat{D}}_N] = \lambda^2\hat{\sigma}^2/N$. Minimising the quadratic in λ at $N \to \infty$ gives $\lambda^\star = \rho\sigma/\hat{\sigma}$ and the value $\sigma^2(1-\rho^2)/n$. □

Proposition 2. Given $\mathcal{G}_t$ (the training half and its decisions), the validation queries are an i.i.d. sample from P independent of $\mathcal{G}_t$. For the human-only bound this is the usual t bound. For the PPI bound, within each fold λ is estimated on the other fold, hence independent of the fold it is applied to, and the unlabeled mean is taken outside B_t; conditionally on λ, $\mathbb{E}[D_j - \lambda\hat{D}_j] = \mu_j - \mu_{\hat{m}_t} - \lambda\mathbb{E}[\hat{D}_j]$ and $\mathbb{E}[\lambda\overline{\hat{D}}_N] = \lambda\mathbb{E}[\hat{D}_j]$, so the estimator is conditionally unbiased for any f, with the variance of Proposition 1. If ACT_t is wrong then for $j^\star = \arg\max_j \mu_j$, $\mu_{j^\star} - \mu_{\hat{m}_t} > \varepsilon \geq U_{t,j^\star}$, i.e. the bound for $(t, j^\star)$ failed; $\hat{m}_t$ is $\mathcal{G}_t$-measurable, so $(t, j^\star)$ is one of the $L|\mathcal{M}|(|\mathcal{M}|-1)$ ordered pairs covered by the union bound, each failing with conditional probability at most α'. □

Proposition 3. Documents in $R_a \cap R_b$ contribute $y_d/Z_a - y_d/Z_b$, documents in $R_b \setminus R_a$ contribute $-y_d/Z_b$, and documents outside $R_a \cup R_b$ contribute nothing. □

C Reproducibility notes

All numbers are regenerated from row-level result files by the scripts in the repository. Each method uses an independent random stream derived from the repeat index and the method name. Batched causal-LM inference with left padding must pass explicit position ids; without them the relevance judgments on one collection (TREC-COVID) collapsed to a constant, which we detected from a 49/50-query anomaly, fixed, and re-ran for every collection (label changes $\leq 0.5\%$ elsewhere; reranker scores unaffected, checked against batch-size-1 scoring). The BCa lower bound on ρ used by the adoption rule covered its target at 0.88–0.92 for pilots of 30 and 50 queries on DBpedia-Entity (0.86 at 20), against 0.75–0.84 for the Fisher-z bound on skewed pairs. The lock document for Section 7 was committed before the target data were accessed; the only pre-lock access was the count of judged queries and pairs.